Group 29 Progress Report: Early Prediction of Student Dropout and Academic Success

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1 Introduction

Universities usually learn that a student is struggling only after the student has withdrawn or failed enough courses that recovery is unrealistic. Dropout is costly for students (debt without a degree), for institutions (lost tuition and weaker graduation statistics) and for society. Advisors cannot manually monitor thousands of students, so an automated early-warning system could let staff offer tutoring, financial aid or counselling while it still matters.

We build such a system as a three-class classifier (Dropout, Enrolled, Graduate) on tabular student records. Our central question is *how early* a student's outcome can be predicted reliably. We ask the same question at three points in time with progressively more information: at enrollment (Stage 1), after semester 1 (Stage 2) and after semester 2 (Stage 3). A moderately accurate Stage 1 model may be more useful than a more accurate Stage 3 model, because only an early prediction leaves time to help.

This report describes our preliminary implementation: the dataset and preprocessing, leakage-safe stage-wise feature sets, a majority-class baseline and three learned models (logistic regression, random forest, gradient boosting), and an evaluation of all of them at all three stages. Challenges we anticipated, and discuss below, are class imbalance, the ambiguity of the "Enrolled" label, weak early signal, feature leakage across stages, and fairness.

2 Related Work

Student retention has long been studied in education research; Tinto (1975) gave an influential theoretical account of dropout from higher education based on academic and social integration. More recent work uses machine learning on institutional records. Our dataset comes from Martins et al. (2021), who built it from several databases of a Portuguese higher-education institution for early prediction of student performance, and is documented in more detail by Realinho et al. (2022). These papers motivate the task and provide the data. Our project differs in that we explicitly build one feature set per prediction point, with variables assigned conservatively to avoid leakage, and compare models across those stages.

For tabular data with mixed numeric and categorical features, tree ensembles are strong default choices. Random forests (Breiman, 2001) average many decorrelated trees, and gradient boosting (Friedman, 2001) builds trees sequentially to correct earlier errors. Efficient implementations such as XGBoost (Chen and Guestrin, 2016) and histogram-based boosting (Ke et al., 2017) are widely used, and Grinsztajn et al. (2022) found that tree-based models often remain ahead of deep learning on medium-sized tabular datasets. We therefore use ensembles of trees, with logistic regression as an interpretable baseline.

Because our classes are imbalanced, we also draw on work on imbalance and probability quality. Chawla et al. (2002) proposed SMOTE oversampling, which we plan to compare against class weighting. Niculescu-Mizil and Caruana (2005) showed that predicted probabilities from many classifiers are poorly calibrated, motivating our planned calibration checks. For fairness, Hardt et al. (2016) define group-level error criteria such as equality of opportunity, which we will use to compare error rates across groups, and Lundberg and Lee (2017) provide SHAP, one option for interpreting predictions.

3 Dataset

We use the public UCI "Predict Students' Dropout and Academic Success" dataset (CC BY 4.0), which contains 4,424 student records with 36 features and one target label (Dropout, Enrolled or

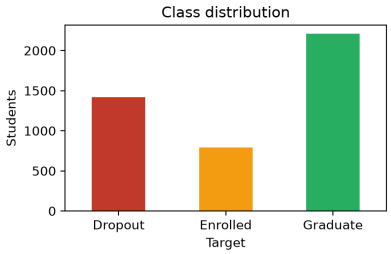


Figure 1: Class distribution of the 4,424 students.

Graduate) drawn from the institution's records, so no manual annotation was needed. Features cover demographics, application and enrollment information, previous education, parents' education and occupation, scholarship and payment status, first- and second-semester curricular unit statistics, and three macroeconomic indicators (unemployment, inflation, GDP). The data are included with our submission (`data.csv`).

Class distribution The classes are imbalanced: Graduate is 49.9% of students (2,209), Dropout 32.1% (1,421) and Enrolled 17.9% (794) (Figure 1). A majority-class predictor therefore reaches about 50% accuracy but only 0.222 macro F1.

Preprocessing The dataset documentation reports that anomalies and missing values were handled by its creators; we verified that there are no missing values and checked for duplicate rows. We stripped stray whitespace from column names, mapped the target to integers (Dropout=0, Enrolled=1, Graduate=2), and dropped the Nationality column from the main feature set because of fairness concerns. Eight nominal code columns (marital status, application mode, course, previous qualification, and the mothers' and fathers' qualification and occupation) are treated as categorical, since their integer codes carry no numeric meaning. All other columns are used as numeric or binary values.

Split We make one stratified 80/20 split (3,539 training and 885 test students) with a fixed random seed. The same split is reused for every stage and model so that stage comparisons are fair. The test set is used only to report final numbers.

4 Features

Stage-wise feature sets We assign each variable to the earliest stage at which it would be known (Table 1). Stage 1 contains demographic, application,

Stage	Adds	# features
1	enrollment, background, macro	21
2	payment status, semester 1	29
3	semester 2	35

Table 1: Feature sets per stage (Nationality excluded).

prior-education, parental, scholarship and macroeconomic variables. Stage 2 adds first-semester curricular unit statistics (credited, enrolled, evaluated, approved, grade, without evaluations) and the payment variables Tuition fees up to date and Debtor, which we place in Stage 2 because payment status can change after enrollment and may already reflect a student's decision to leave. Stage 3 adds the second-semester statistics. Scholarship status stays in Stage 1 because scholarships are generally awarded as part of enrollment, and macroeconomic indicators stay in Stage 1 because they do not depend on the student. Each model only receives its own stage's features, which prevents leakage from later semesters into earlier predictions.

Representation For logistic regression we one-hot encode the eight nominal columns and standardize the numeric columns. For tree models we ordinal-encode the nominal columns and leave numeric columns unscaled, since trees are insensitive to monotone rescaling. We have not yet done any hand-crafted feature engineering or automatic feature selection; the only selection is the stage grouping and the removal of Nationality. Ratio features (for example units approved divided by units enrolled) and a semester-1 to semester-2 grade change are candidates for future work.

5 Implementation

All models are implemented with scikit-learn (Pedregosa et al., 2011), each as a pipeline of preprocessing and classifier.

Baselines and models The required baseline is a majority-class predictor (always Graduate). We train three learned models: (1) multinomial logistic regression, an interpretable baseline; (2) a random forest; and (3) histogram-based gradient boosting (scikit-learn's `HistGradientBoostingClassifier`), the main model, which we expect to be strongest on tabular data.

Loss functions and optimization Logistic regression minimizes the L2-regularized multinomial

cross-entropy

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} w_{y_i} \log p_{i,y_i} + \tfrac{1}{2C}\|\theta\|_2^2,$$

optimized with L-BFGS. Gradient boosting minimizes the same multiclass cross-entropy by adding one tree per iteration along the negative gradient of the loss, with a 10% internal validation split for early stopping. Random forest has no global loss; each tree chooses splits by Gini impurity and predictions are averaged over trees.

Class imbalance All three learned models use balanced class weights, $w_c = N/(KN_c)$ with $K = 3$, so that errors on Dropout and Enrolled are penalized more heavily than errors on Graduate.

Hyperparameter tuning For each stage and model we run randomized search with 5-fold stratified cross-validation on the training set, scoring macro F1, and then refit the best configuration on the full training set. Search spaces: regularization strength C (log-uniform in $[10^{-3}, 10^{2}]$) for logistic regression; number of trees, maximum depth, minimum leaf size and `max_features` for the random forest; and learning rate, depth, number of leaves, minimum leaf size, L2 penalty and iterations for boosting. Selected values are saved in `best_params.json`.

Ablations So far, the stage comparison is itself our main ablation: it measures what each additional group of features (semester 1, then semester 2) contributes to every model. We have not yet ablated imbalance handling (class weights versus SMOTE versus none) or the macroeconomic features.

6 Results and Evaluation

Evaluation strategy We use the single stratified train/test split from Section 3. Validation estimates come from 5-fold stratified cross-validated out-of-fold predictions on the training set; test estimates come from the refit model on the held-out test set. Macro F1 and dropout recall are our main metrics, because accuracy is misleading with imbalanced classes and a missed dropout is the costliest error. We also report dropout precision, Enrolled recall, balanced accuracy, log loss and the multiclass Brier score, and examine confusion matrices.

Findings Table 2 summarizes results. (i) All learned models far exceed the majority baseline (0.222 macro F1, zero dropout recall). (ii) Waiting helps, mostly once. The best test macro F1 rises from about 0.52 at Stage 1 to about 0.68 at Stage 2 and 0.71 at Stage 3, so most of the gain arrives with semester-1 information, while semester 2 adds comparatively little. (iii) At every stage the three learned models are within about 0.02 test macro F1 of each other, which on 885 test students is too close to declare a winner; validation scores are similarly close, with gradient boosting marginally ahead. Logistic regression is surprisingly competitive, suggesting much of the signal is close to linear in the semester performance features. (iv) Enrolled is the hardest class (recall 0.35–0.47 at Stage 1, 0.56–0.69 at Stage 3), consistent with its unresolved outcome. (v) Validation and test macro F1 differ by at most about 0.04, so we are not seeing severe overfitting to the tuning folds, although validation scores are slightly optimistic because hyperparameters were chosen on the same folds. (vi) Test log loss improves from about 0.94–0.97 at Stage 1 to about 0.60–0.63 at Stage 3.

7 Feedback and Plans

Reflection Our results show that Stage 1 is weak (macro F1 about 0.5), that the learned models are nearly tied, and that Enrolled is the main source of error. Because the models are so close, we cannot yet claim that boosting is better than a regularized linear model, and our single split on 885 test students gives noisy comparisons.

Plans (1) Reduce noise in comparisons by repeating cross-validation with several seeds and reporting confidence intervals, and test whether model differences are significant. (2) Improve Enrolled and dropout recall with threshold tuning on predicted probabilities and compare class weighting against SMOTE (Chawla et al., 2002). (3) Add engineered features such as approval ratios and semester-to-semester changes, and ablate the macroeconomic features. (4) Check calibration of predicted probabilities and calibrate if needed. (5) Interpret the best models with permutation importance and SHAP. (6) Run fairness checks (for example by gender and scholarship status, using group-wise recall) and test whether features such as International status, Displaced or parental variables act as proxies for nationality; if time permits, run the secondary experiment including Nationality.

Stage	Model	Val F1	Test F1	Bal. Acc.	Dropout R	Dropout P	Enrolled R
—	Majority	.222	.222	.333	.000	.000	.000
1	LogReg	.533	.520	.531	.563	.604	.465
1	RandomForest	.537	.494	.499	.553	.520	.346
1	GradBoost	.538	.510	.519	.542	.572	.434
2	LogReg	.682	.664	.673	.658	.803	.585
2	RandomForest	.691	.684	.686	.701	.787	.528
2	GradBoost	.696	.679	.691	.669	.812	.629
3	LogReg	.716	.701	.715	.683	.822	.667
3	RandomForest	.721	.715	.715	.718	.826	.560
3	GradBoost	.724	.710	.726	.704	.840	.692

Table 2: Macro F1 (validation = cross-validated, and test), and test balanced accuracy, dropout recall (R), dropout precision (P) and Enrolled recall. The majority baseline always predicts Graduate and is identical at all stages.

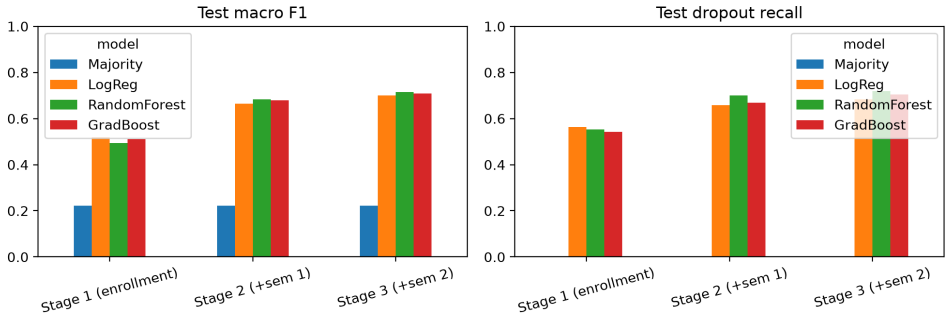


Figure 2: Test macro F1 and dropout recall by stage and model.

(7) If the three-class task proves too overlapping for the Enrolled class, discuss it as a limitation rather than switching to a binary task, following the TA's advice.

Team Contributions

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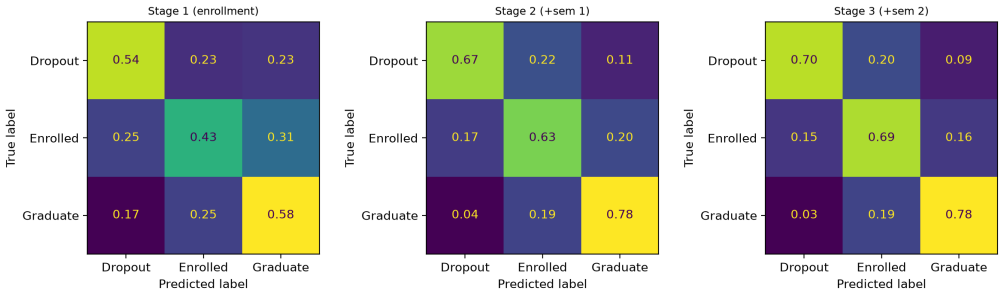


Figure 3: Row-normalized test confusion matrices for gradient boosting at each stage.

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